

OCR学习笔记

1、进入百度云，然后登陆，点击控制台，然后搜文字识别，新建应用

百度智能云-云智一体深入产业



搜OCR文字识别



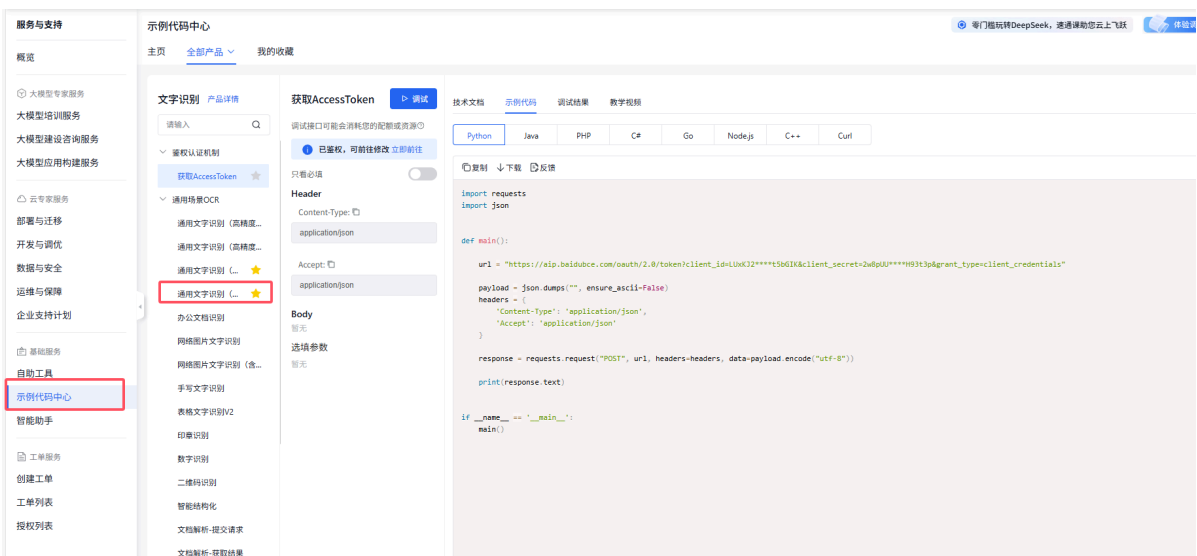
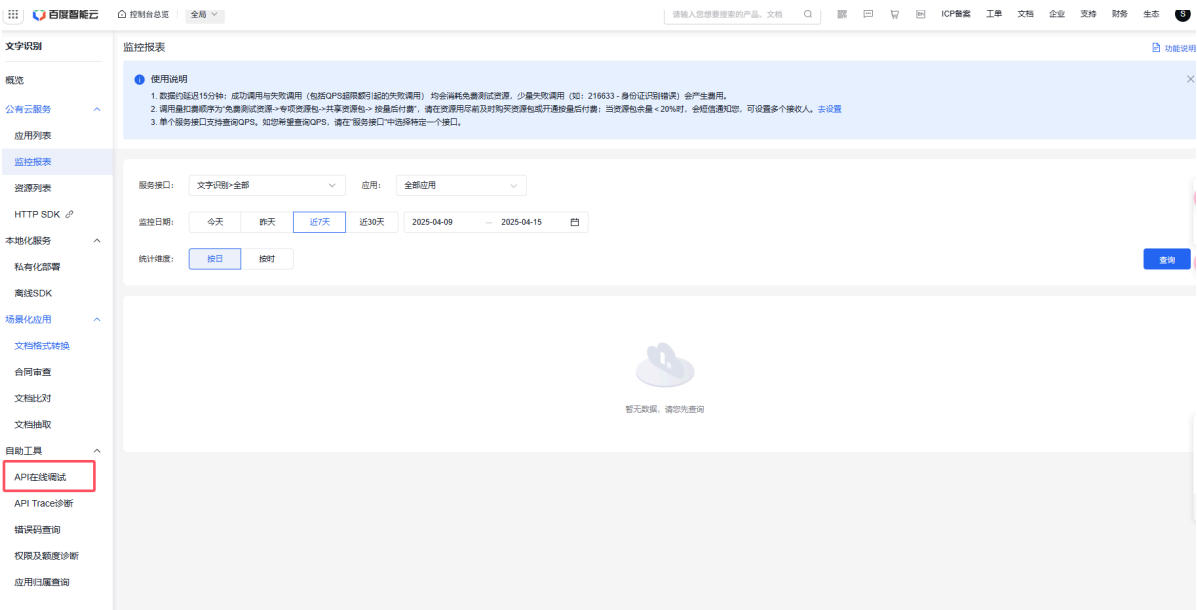
新建应用

创建应用 ☐ 显示全部应用							
序号	应用名称	AppID	API Key	Secret Key	创建时间	创建人	操作
1	python_orc	118494985	LUxKJ... 展开 复制	Zw8pU... 展开 复制	2025-04-14 22:34:03	sweetheartree	报表 管理 删除

接口需要存在可用资源才能正常调用, [查看可用资源](#)

Secret Key: 2w8pUU3Zt7tkQTH8GVKGQLSOEUH93t3p [复制](#)

- ```
1 LuxKJ2sTQu41tj2eOmt5bGIK
2
3 2w8pUU3Zt7tkQTH8GVKGQLSOEUH93t3p
4
5 6715044
```



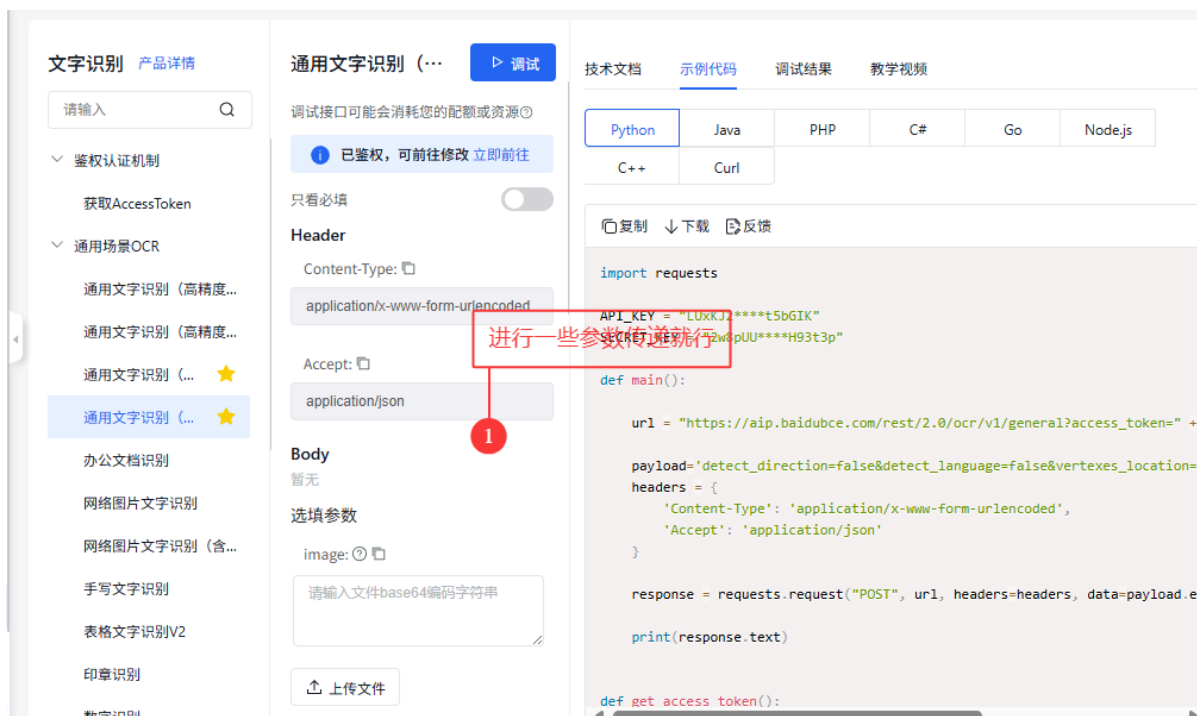
# Postman



## 5、2 直接在线调试就行



## 6、可以进行在线的代码调试



## 7、代码如下，这里用了模糊匹配和文字帧率的显示，传坐标值

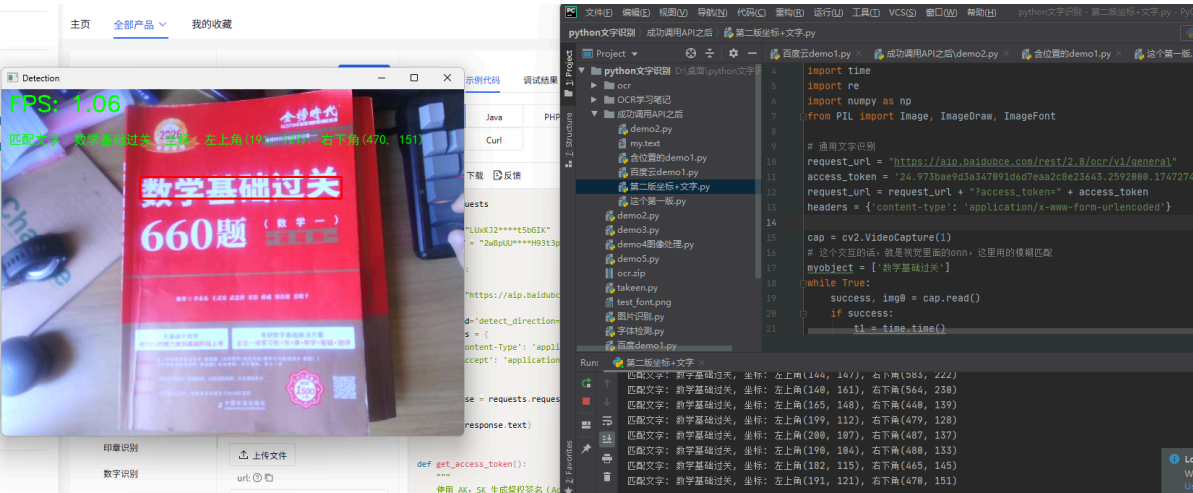
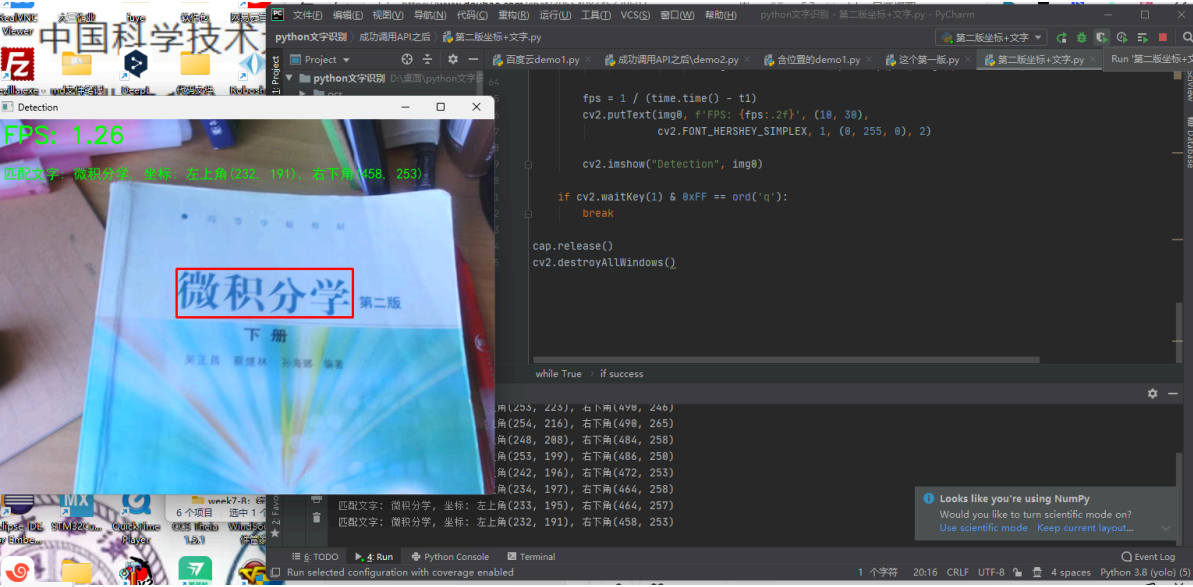
```
1 import requests
2 import base64
3 import cv2
4 import time
5 import re
6 import numpy as np
7 from PIL import Image, ImageDraw, ImageFont
8
9 # 通用文字识别
10 request_url = "https://aip.baidubce.com/rest/2.0/ocr/v1/general"
11 access_token =
12 '24.973bae9d3a347091d6d7eaa2c8e23643.2592000.1747274046.282335-118494985'
13 request_url = request_url + "?access_token=" + access_token
14 headers = {'content-type': 'application/x-www-form-urlencoded'}
15
16 cap = cv2.VideoCapture(1)
17 # 这个交互的话，就是视觉里面的onn，这里用的模糊匹配
18 myobject = ['微积分']
19 while True:
20 success, img0 = cap.read()
21 if success:
22 t1 = time.time()
23 # 将图像转换为 Base64 编码
24 _, buffer = cv2.imencode('.png', img0)
25 img_base64 = base64.b64encode(buffer)
26
27 # 添加额外的请求参数
28 params = {
29 "image": img_base64,
30 "language_type": "CHN_ENG",
31 "detect_direction": "false",
```

```

31 "detect_language": "false",
32 "vertexes_location": "true",
33 "paragraph": "true",
34 "probability": "true"
35 }
36
37 # 调用百度 OCR API
38 response = requests.post(request_url, data=params, headers=headers)
39 if response:
40 result = response.json()
41 if 'words_result' in result:
42 for item in result['words_result']:
43 text = item['words']
44 for obj in myobject:
45 if re.search(obj, text):
46 vertexes = item['vertexes_location']
47 x1 = vertexes[0]['x']
48 y1 = vertexes[0]['y']
49 x2 = vertexes[2]['x']
50 y2 = vertexes[2]['y']
51 cv2.rectangle(img0, (x1, y1), (x2, y2), (0, 0,
255), 2)
52
53 print(f"匹配文字: {text}, 坐标: 左上角({x1}, {y1}),
54 右下角({x2}, {y2})")
55
56 # 在窗口左上角显示匹配文字和坐标
57 # 强制类型转换
58 info_text = f"匹配文字: {str(text)}, 坐标: 左上角
59 ({str(x1)}, {str(y1)}), 右下角({str(x2)}, {str(y2)})"
60
61 # 将 OpenCV 图像转换为 PIL 图像
62 pil_image = Image.fromarray(cv2.cvtColor(img0,
63 cv2.COLOR_BGR2RGB))
64
65 draw = ImageDraw.Draw(pil_image)
66 # 加载中文字体文件, 这里使用的是黑体, 你可以根据需要更换
67 font = ImageFont.truetype("simhei.ttf", 18)
68 draw.text((10, 60), info_text, font=font, fill=
69 (0, 255, 0))
70
71 # 将 PIL 图像转换回 OpenCV 图像
72 img0 = cv2.cvtColor(np.array(pil_image),
73 cv2.COLOR_RGB2BGR)
74
75 fps = 1 / (time.time() - t1)
76 cv2.putText(img0, f'FPS: {fps:.2f}', (10, 30),
77 cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)
78
79 cv2.imshow("Detection", img0)
80
81 if cv2.waitKey(1) & 0xFF == ord('q'):
82 break
83
84 cap.release()
85 cv2.destroyAllWindows()

```

# 8、结果如下



# 9、注意事项

这里调用之后记得关掉, 免费的有额度, 这里每个月有免费的额度, 付费的话, 50块1万次吧

